Journal of Infection 92 (2026) 106753

Contents lists available at ScienceDirect

Journal of Infection

journal homepage: www.elsevier.com/locate/jinf

Infectious Disease Practice

Effectiveness of bivalent COVID-19 vaccines against SARS-CoV-2 reinfection in patients with cancer: Evidence from a nationwide target trial emulation

Hyun-Kyung Kim [a], Suyoung Jo [b,*,1], Kyung-Duk Min [c], Sung-il Cho [a,b,**,1]

[a] Graduate School of Public Health, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea
[b] Institute of Health and Environment, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea
[c] College of Veterinary Medicine, Chungbuk National University, 1 Chungdae-ro, Seowon-Gu, Cheongju, Chungbuk 28644, Republic of Korea

ARTICLE INFO

Article history:
Accepted 20 April 2026
Available online 28 April 2026

Keywords:
COVID-19
Vaccine effectiveness
Cancer
Target trial emulation
Cohort study

SUMMARY

Objectives: Cancer patients are vulnerable to SARS-CoV-2 reinfection due to impaired immunity. We estimated the effectiveness of bivalent COVID-19 vaccination against reinfection in adults with cancer and prior SARS-CoV-2 infection.

Methods: We conducted a nationwide target trial emulation study using the K-CoV-N cohort, including 84,748 adults with cancer and prior SARS-CoV-2 infection (follow-up: Oct 2022–Jun 2023). The clone-censor-weight method with inverse probability of censoring weights compared bivalent vaccination versus none. SARS-CoV-2 reinfection ≥14 days post-vaccination and ≥90 days after prior infection was the primary outcome. Vaccine effectiveness (VE) was estimated via pooled logistic regression (1-HR).

Results: 24.1% received bivalent vaccines. Reinfection risk was 5.2% (bivalent) versus 8.4% (no-bivalent). Overall, VE was 22.2% (95% CI, 15.8–28.2%). Protection was greater among younger adults, those with prior boosters, and those with more distant prior infections (≥1 year). Significant protection was observed in patients with solid tumors, whereas estimates for hematologic malignancies were attenuated and non-significant.

Conclusions: Bivalent vaccination meaningfully reduced reinfection risk among adults with cancer, extending and strengthening the existing evidence base through a causal inference framework. These findings provide real-world evidence to support their continued role in routine and seasonal immunization strategies while highlighting the need for risk-stratified approaches tailored to immune capacity.

© 2026 The Author(s). Published by Elsevier Ltd on behalf of British Infection Association. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Introduction

Should cancer patients continue to receive seasonal coronavirus disease 2019 (COVID-19) vaccinations? Cancer patients are universally prioritized for COVID-19 vaccination in national immunization programs worldwide,[1–3] yet a fundamental paradox underlies this policy: despite their clear vulnerability to severe COVID-19 outcomes, evidence on whether currently recommended seasonal vaccination strategies meaningfully reduce SARS-CoV-2 infection in this population has not been adequately estimated at the population level under a modern causal inference framework. This evidence-policy gap is clinically consequential given the high stakes in this vulnerable group.

Cancer patients face a sustained vulnerability to severe outcomes from COVID-19, due to malignancy- and treatment-related immunosuppression,[4] resulting in high risks of hospitalization and mortality.[5–7] Beyond acute illness, COVID-19 poses a threat to treatment continuity, as infection-related delays in anticancer therapy can compromise survival outcomes.[8–10] Compounding this vulnerability, vaccine-induced protection in cancer patients is both attenuated and short-lived, with antibody responses waning within 3 months compared to 6 months in the general population.[11] As a result, SARS-CoV-2 reinfection represents a persistent and clinically significant threat in this population. Although prior infection confers some degree of protection,[12] reinfection risks remain elevated despite widespread vaccination and prior infection among immunocompromised individuals.[13,14]

* Corresponding author.
** Corresponding author at: Graduate School of Public Health, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea.
E-mail addresses: kimhk1878@snu.ac.kr, hyunkyungkim929@gmail.com (H.-K. Kim), josuyoung1@gmail.com (S. Jo), kdmin@cbnu.ac.kr (K.-D. Min), persontime@hotmail.com (S.-i. Cho).
[1] These authors contributed equally.

https://doi.org/10.1016/j.jinf.2026.106753
0163-4453/© 2026 The Author(s). Published by Elsevier Ltd on behalf of British Infection Association. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

To counter immune evasion by emerging variants and accelerated waning, major public health authorities including the World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) have recommended variant-updated vaccines for high-risk populations.[15,16] Consequently, bivalent vaccines targeting both ancestral and Omicron strains were introduced in late 2022 as the standard booster recommendation globally,[17] with South Korea launching its national bivalent vaccination program in October 2022, which explicitly prioritized immunocompromised individuals including cancer patients as high-risk groups. Yet despite the broad recommendations for repeated boosting in cancer patients, the effectiveness of bivalent vaccines against reinfection in this population has not been well characterized. As randomized trials have largely excluded immunocompromised individuals,[18] observational studies remain the primary source of evidence in this setting.

Furthermore, studies of COVID-19 vaccine effectiveness have largely focused on primary infection or severe outcomes[19–22] rather than reinfection, which has become the dominant transmission mode in the endemic phase. Among reinfection studies, most have been conducted in the general population[23–26] or across diverse immunocompromising conditions,[27] limiting their applicability to cancer patients specifically. Crucially, cancer-specific estimates of bivalent vaccine effectiveness against reinfection remain scarce. While existing cancer-specific studies have contributed valuable insights,[28–30] many have been constrained by methodological challenges inherent to conventional observational designs, including confounding by indication and inadequate handling of time-varying exposures,[31,32] which may limit the precision of effectiveness estimates. Complementing this evidence base with approaches that more explicitly address such sources of bias therefore remains important, particularly as countries including South Korea integrate COVID-19 vaccination into national immunization programs,[33,34] where policy decisions increasingly require cancer-specific effectiveness data.

To address this gap, we conducted a nationwide target trial emulation (TTE) to estimate the effect of bivalent COVID-19 vaccination on SARS-CoV-2 reinfection risk among adults with cancer in South Korea, with follow-up from October 4, 2022, through June 30, 2023. We used the K-CoV-N (Korea Disease Control and Prevention Agency (KDCA)-COVID19-National Health Insurance Service (NHIS)) cohort, a nationwide, claims-based cohort that integrates all laboratory-confirmed infections in South Korea. The NHIS operates South Korea's mandatory single-payer health insurance program, covering more than 97% of the national population.[35,36] By emulating a hypothetical randomized trial under the TTE framework, we aimed to provide real-world evidence on bivalent vaccine effectiveness in cancer patients under a causal inference framework. These findings extend the existing evidence base and directly inform seasonal vaccination policy for this highly vulnerable population as COVID-19 transitions to endemic circulation.

Methods

Data sources

Data were obtained from the K-CoV-N cohort, a comprehensive database established by the KDCA and NHIS to track COVID-19 infections and vaccination status across the entire Korean population.[37] The KDCA dataset includes all laboratory-confirmed SARS-CoV-2 infections reported nationwide, including dates of infection, dates of vaccination, and vaccine types for each confirmed case. Between October 7, 2020, and June 30, 2023, a total of 32,187,621 individuals in South Korea were confirmed to have COVID-19, with approximately 97.8% of cases occurring during the Omicron-dominant period. Although infections during this period overwhelmingly dominated the epidemic, cases from the Delta and early transitional phases provide essential contrast for assessing temporal variation in reinfection dynamics and vaccine effectiveness. To ensure sufficient statistical precision and comparability across epidemic contexts, we therefore applied stratified random sampling, selecting individuals such that infections from the Delta- and Omicron-dominant periods were included at a 2:3 ratio. The resulting sample preserved the national age-, sex-, and region-specific distributions of confirmed cases.

For all individuals captured in the KDCA dataset, comprehensive linkage to the NHIS database was performed using unique personal identifiers. The NHIS database (2018–2023) contains deidentified information on demographic characteristics (age, sex, disability status, area of residence, etc.), date of death, and comprehensive medical claims including diagnoses coded using the International Classification of Diseases, 10th Revision (ICD-10) and procedures.[36] This deterministic linkage using unique personal identifiers enabled complete longitudinal tracking of vaccination status, SARS-CoV-2 infection history, cancer diagnoses, treatment patterns and clinical outcomes for all individuals.

Study population and eligibility criteria

The study population included eligible adults aged 19 years or older with a record of cancer-related healthcare encounters between January 1, 2018, and February 10, 2021, a period selected to identify cancer diagnoses prior to the initiation of COVID-19 vaccination in South Korea (Fig. S1). Cancer was identified through the presence of ICD-10 codes C00–C97 in a primary diagnosis with cancer-specific registration codes (V027, V193, V194) (Table S1).[38,39] To ensure all participants were at risk for reinfection during the bivalent vaccination period, eligibility required SARS-CoV-2 infection prior to the index date of October 4, 2022, when the Korean government launched the national bivalent vaccination program.[40] Participants were excluded if they received bivalent vaccination outside the national immunization program, including overseas vaccination, or if infection was recorded after the date of death (Fig. 1). A total of 84,748 individuals were included.

Study design (target trial emulation framework)

We conducted a nationwide, population-based cohort study emulating a target trial to estimate the effect of bivalent COVID-19 vaccination on SARS-CoV-2 reinfection risk among adults with cancer in South Korea. The TTE approach, a framework for causal inference from observational data,[41] explicitly emulates the design of a hypothetical randomized trial to address time-varying confounding, immortal time bias, and selection bias that commonly challenge conventional observational vaccine studies.[31,32] Specifically, this is achieved by aligning eligibility determination, treatment assignment, and the start of follow-up at a common time zero, preventing the misalignment that gives rise to these biases. The validity of this framework relies on the assumptions of consistency, positivity and conditional exchangeability,[42,43] each of which is assessed in detail in Supplementary text 1. This study adhered to the Transparent Reporting of Observational Studies Emulating a Target Trial (TARGET) reporting guidelines,[44] and the protocol of the hypothetical target trial is detailed in Table S2.

Exposure and outcome

The exposure of interest was receipt of a bivalent COVID-19 vaccine (BNT162b2 bivalent or mRNA-1273.214) between October 4, 2022, and June 30, 2023. Participants were classified by time-varying vaccination status, contributing person-time to either the bivalent vaccination group (after receipt) or the no-vaccination group (before

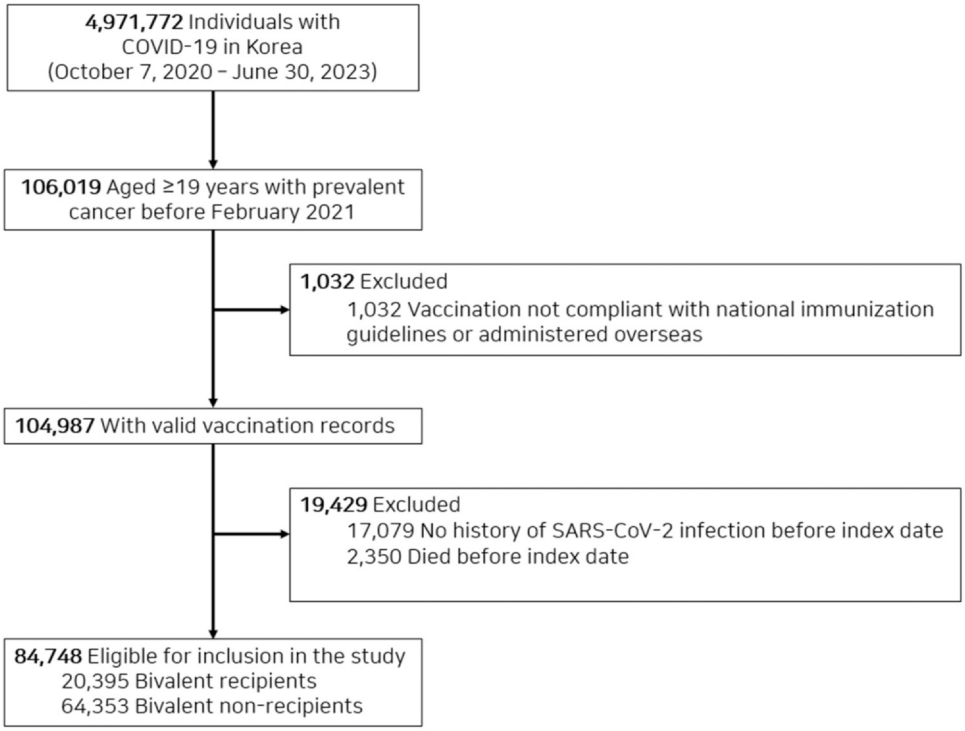

Fig. 1. Flowchart of eligible individuals. Flowchart illustrating the selection of eligible individuals with cancer and prior SARS-CoV-2 infection for inclusion in the study. Among 4,971,772 individuals with COVID-19 in Korea between October 7, 2020, and June 30, 2023, 106,019 adults aged ≥19 years with prevalent cancer before February 2021 were identified. After excluding individuals with noncompliant or overseas vaccination records and those without prior SARS-CoV-2 infection or who died before the index date, 84,748 individuals were included in the final cohort (20,395 bivalent recipients and 64,353 bivalent non-recipients).

bivalent vaccination or if remaining unvaccinated). The primary outcome was SARS-CoV-2 reinfection, defined as a positive PCR at least 14 days after bivalent vaccination[45] and at least 90 days after any prior SARS-CoV-2 infection.[46,47] The 14-day post-vaccination exclusion period allowed for vaccine-induced immunity development,[48] while the 90-day interval between infections minimized misclassification of prolonged viral shedding as reinfection.[49,50] Infections within these windows were excluded from outcome ascertainment.

Covariates

Baseline covariates included demographic factors (age, sex), comorbidity burden quantified by the Charlson Comorbidity Index (CCI) score calculated from ICD-10 codes (Table S3), and COVID-19-related factors including prior SARS-CoV-2 vaccination history (primary series only, one or two monovalent boosters), time since the most recent vaccination, number of prior SARS-CoV-2 infections, and time since the most recent infection, each measured as of the index date. Cancer-related characteristics included cancer type categorized by anatomic site into solid tumors and hematologic malignancies (detailed ICD-10 code classifications provided in Table S1), and chemotherapy status identified through procedure codes (Table S4).

Statistical analysis

The clone-censor-weight (CCW) method was employed to implement the TTE framework, approximating a per-protocol analysis of a randomized trial and enabling estimation of effects under sustained adherence.[51] Each individual was cloned at the index date (October 4, 2022) and assigned to both strategies: (1) receipt of bivalent vaccine (outcome ascertainment starting 14 days post-vaccination to allow for immunity), or (2) no-bivalent vaccination. Clones were censored upon deviation from their assigned protocol. Follow-up ended at the earliest occurrence of SARS-CoV-2 reinfection (the outcome of interest), death, or administrative end (June 30, 2023). This design prevents immortal time bias by ensuring clones contribute person-time only during adherence.[52]

Informative censoring due to nonadherence was addressed using inverse probability of censoring weights (IPCW), estimated from a Cox proportional hazards model fitted at each person-period, adjusting for linear and quadratic terms for time, age, sex, CCI score, number of prior SARS-CoV-2 vaccinations, time since the most recent vaccination, number of prior infections, time since the most recent infection, cancer type, and chemotherapy status. The inverse of the cumulative predicted probability of remaining uncensored was used as the IPCW for each person-time observation, creating a weighted pseudo-population representative of the cohort under full adherence (Fig. 2). Stabilized weights were truncated at the 1st and 99th percentiles.

Using the IPCW-weighted dataset, we first estimated inverse-probability-weighted (IPW) Kaplan–Meier curves to compare cumulative incidence under the two vaccination strategies, accounting for time-varying adherence patterns inherent to the CCW design.[53,54]

We then fitted a weighted pooled logistic regression model to estimate the hazard ratio (HR) for reinfection comparing bivalent vaccination with no-bivalent vaccination. This model accommodates the discrete-time data and the time-varying weights, approximating the parameters of a marginal structural Cox model.[55,56] The model included the same baseline covariates used in the censoring model. Vaccine effectiveness (VE) was calculated as (1 − HR) × 100%, and 95% confidence intervals (CIs) were derived using robust variance estimators accounting for repeated observations from cloned individuals.

Prespecified subgroup analyses examined heterogeneity across age groups (20–49, 50–64, or ≥65 years), sex, cancer type (solid or hematologic malignancies, and specific cancer subtypes), prior

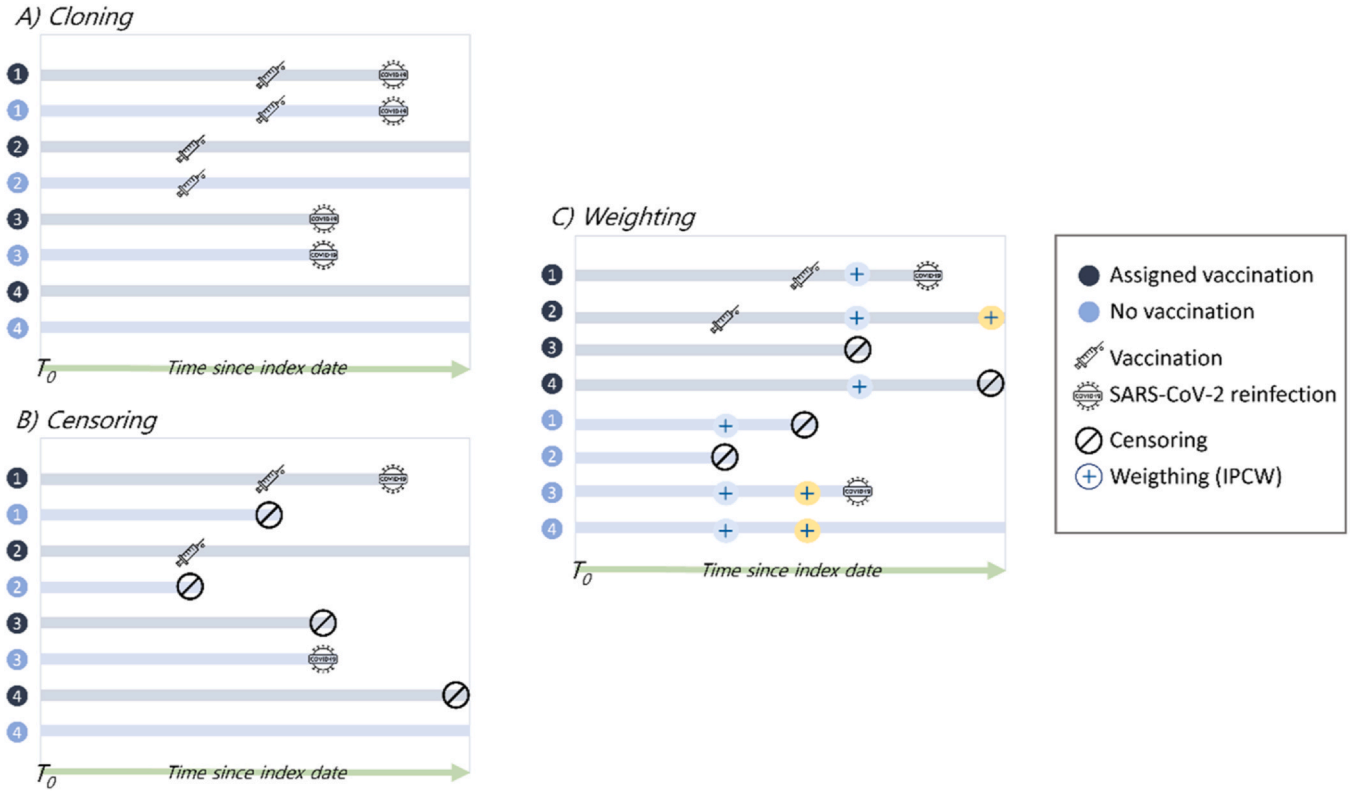

Fig. 2. The clone-censor-weight (CCW) approach within the target trial emulation. The clone–censor–weight approach used in this emulation of the target trial on bivalent COVID-19 vaccination, showing how individuals were cloned to both strategies, censored when deviating from the assigned strategy, and weighted using inverse probability of censoring.

vaccination history (primary series only, one monovalent booster, two or more monovalent boosters), number of prior SARS-CoV-2 infections (1 or ≥2), and chemotherapy status (yes or no). VE estimates and 95% CIs were calculated separately within each subgroup using the same approach.

All analyses were conducted using SAS version 9.4 (SAS Institute), and figures were generated using the statistical software R (version 4.2.3).

Ethical approvals

This study was exempted from review by the Institutional Review Board of Seoul National University (IRB No. E2503/003-003).

Results

General characteristics of the study population

A total of 84,748 adults aged 19 years or older with cancer and prior SARS-CoV-2 infection between October 4, 2022, and June 30, 2023 were included (Fig. 1). Participants contributed 21,368,296 person-days: 3,760,959 person-days under the bivalent vaccination and 17,607,337 person-days under the no-bivalent vaccination. Overall, 20,395 individuals (24.1%) received a bivalent vaccine, of whom 1068 (5.2%) experienced reinfections. A total of 64,353 individuals (75.9%) did not receive a bivalent vaccine, among whom 5391 (8.4%) developed reinfection.

Baseline characteristics differed between groups (Table 1). Bivalent recipients were older (69.8% aged ≥65 years vs. 31.1%), more likely to be male (55.9% vs. 36.5%), and had a higher CCI score ≥3 (32.3% vs. 22.6%). Prior vaccination history differed substantially: 59.9% of bivalent recipients had received two or more boosters versus 17.6% of non-recipients, whereas only 4.7% of recipients had received the primary series alone versus 20.5% of non-recipients. Prior infection history was comparable: 96.9% of recipients and 96.7% of non-recipients had a single prior infection, and mean time since the most recent infection was similar (188 vs. 187 days).

Cancer subtype distribution varied: colorectal (11.3%) and non-colorectal gastrointestinal cancers were more common among recipients, while endocrine (25.6%) and breast cancer (15.5%) were more common among non-recipients. Hematologic malignancies and chemotherapy receipt were similar between groups (4.1% vs. 4.9% and 4.5% vs. 5.1%).

Cumulative incidence curves demonstrated early and persistent divergence, with clear separation evident within 30 days (Fig. 3). By 180 days, the cumulative incidence of reinfection was 667 (95% CI, 606–728) per 100,000 person-days under the bivalent vaccination and 4614 (95% CI, 4486–4741) per 100,000 person-days under the no-bivalent vaccination.

Vaccine effectiveness against SARS-CoV-2 reinfection

Under the TTE framework and its stated assumptions, bivalent vaccination was associated with a 22.2% reduction (95% CI, 15.8–28.2%) in the risk of SARS-CoV-2 reinfection (Fig. 4). Reinfection incidence was lower with bivalent (69.36 per 1000 person-years) versus no-bivalent vaccination (93.36 per 1000 person-years).

VE varied significantly across clinical subgroups (Fig. 4). Older adults—who constituted the majority of bivalent recipients—showed a lower VE compared to younger patients (18.9%; 95% CI, 10.5–26.4% for ages ≥65 years vs. 40.3%; 95% CI, 18.4–56.3% for ages 20–49 years). Individuals with prior monovalent boosters showed clear protection (25.2% for 1 booster; 21.1% for 2 boosters) compared to primary series only (14.4%; 95% CI, −17.6% to 37.7%), with similar VE between 1 and 2 prior boosters. Individuals with ≥2 prior infections showed negative VE (−49.8%; 95% CI, −130.6% to 2.7%) compared to those with a single prior

Table 1
Baseline characteristics of cancer patients by bivalent COVID-19 vaccination status.

Characteristic	Participants, No. (%)		
	Bivalent recipients	Bivalent non-recipients	Overall
All	20,395 (100.0)	64,353 (100.0)	84,748 (100.0)
Age, mean (SD), y	69.03 (11.5)	57.45 (14.0)	60.24 (14.3)
Age, y			
20–49	1269 (6.2)	19,217 (29.9)	20,486 (24.2)
50–64	4885 (24.0)	25,143 (39.1)	30,028 (35.4)
≥65	14,241 (69.8)	19,993 (31.1)	34,234 (40.4)
Sex			
Female	8987 (44.1)	40,846 (63.5)	49,833 (58.8)
Male	11,408 (55.9)	23,507 (36.5)	34,915 (41.2)
Prior vaccination history			
No	0 (0.00)	5384 (8.37)	5384 (6.4)
Primary series only	968 (4.8)	13,202 (20.5)	14,170 (16.7)
Booster one	7201 (35.3)	34,441 (53.5)	41,642 (49.1)
Booster two	12,226 (60.0)	11,326 (17.6)	23,552 (27.8)
Time since the most recent vaccination, mean (SD), days	199.57 (94.5)	262.89 (92.8)	246.61 (97.3)
Time since the most recent vaccination, days			
< 90	3415 (16.7)	5975 (9.3)	9390 (11.1)
< 180	7245 (35.5)	5039 (7.8)	12,284 (14.5)
< 1 year	9325 (45.7)	42,873 (66.6)	52,198 (61.6)
≥1 year	410 (2.0)	5082 (7.9)	5492 (6.5)
No prior vaccination	0 (0.00)	5384 (8.4)	5384 (6.4)
Number of previous SARS-CoV2 infection			
1	19,769 (96.9)	62,226 (96.7)	81,995 (96.8)
≥2	626 (3.1)	2127 (3.3)	2753 (3.3)
Time since the most recent infection, mean (SD), days	188.13 (98.7)	187.08 (98.3)	187.33 (98.4)
Time since the most recent infection, days			
< 90	4090 (20.1)	13,600 (21.1)	17,690 (20.9)
< 180	2471 (12.1)	6575 (10.2)	9046 (10.7)
< 1 year	13,112 (64.3)	41,865 (65.1)	54,977 (64.9)
≥1 year	722 (3.5)	2313 (3.6)	3035 (3.6)
CCI score			
0–2	13,816 (67.7)	49,793 (77.4)	63,609 (75.1)
≥3	6579 (32.3)	14,560 (22.6)	21,139 (24.9)
Cancer type			
Solid	19,713 (96.7)	61,630 (95.8)	81,343 (95.3)
Hematologic	834 (4.1)	3141 (4.9)	3975 (4.7)
Cancer subtype			
Lip, oral cavity, pharynx	340 (1.4)	940 (1.3)	1280 (1.3)
Colorectal	2667 (11.3)	6507 (8.8)	9174 (9.4)
Non-colorectal gastrointestinal	5078 (21.6)	11,070 (15.0)	16,148 (16.6)
Lung	1711 (7.3)	3420 (4.6)	5131 (5.3)
Respiratory/intrathoracic	264 (1.1)	713 (1.0)	977 (1.0)
Bone/mesothelial/soft tissue	982 (4.2)	2274 (3.1)	3256 (3.4)
Breast	2202 (9.3)	11,442 (15.5)	13,644 (14.0)
Gynecological	693 (2.9)	4093 (5.6)	4786 (4.9)
Prostate	2754 (11.7)	3258 (4.4)	6012 (6.2)
Male urological	31 (0.1)	191 (0.3)	222 (0.2)
Urinary tract	1614 (6.9)	3347 (4.6)	4961 (5.1)
CNS	183 (0.8)	806 (1.1)	989 (1.0)
Endocrine glands	3318 (14.1)	18,831 (25.6)	22,149 (22.8)
Lymphoma	418 (1.8)	1442 (2.0)	1860 (1.9)
Myeloma	262 (1.1)	923 (1.3)	1185 (1.2)
Leukemia	223 (1.0)	1006 (1.4)	1229 (1.2)
Others	826 (3.5)	3377 (4.6)	4203 (4.3)
Chemotherapy			
No	19,481 (95.5)	61,043 (94.9)	80,524 (95.0)
Yes	914 (4.5)	3310 (5.1)	4224 (5.0)

Abbreviations: SD, standard deviation; CCI, Charlson Comorbidity Index; CNS, central nervous system; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.
Notes. Unless otherwise indicated, data are presented as number (percentage) of participants. Time since the most recent vaccination and time since the most recent infection were calculated relative to the index date.

infection (23.9%; 95% CI, 17.5–29.9%). A similar pattern was observed when stratifying by time since the most recent infection: individuals infected within the past 90 days showed no measurable benefit from bivalent vaccination, whereas VE increased progressively for intervals of 3–6 months (21.7%; 95% CI, 0.3–38.5%), < 1 year (26.8%; 95% CI, 19.9–33.0%), and ≥1 year (35.8%; 95% CI, 10.1–54.2%) after infection.

Effectiveness varied across cancer types. Solid tumors overall experienced measurable protection, with an adjusted VE of 21.9% (95% CI, 15.3–27.9%), whereas hematologic malignancies had attenuated and imprecise estimates (14.8%; 95% CI, −21.7% to 40.3%), indicating substantial uncertainty after stratification. Among solid tumor subtypes, colorectal (32.9%; 95% CI, 15.7–46.6%), non-colorectal gastrointestinal (18.1%; 95% CI, 3.0–30.8%), lung (35.7%; 95% CI, 15.6–51.0%), and endocrine cancers (24.0%; 95% CI, 8.2–37.1%) showed statistically robust protection. For the remaining subtypes—including breast, gynecologic, urinary tract, male urological, lip/oral/pharyngeal, respiratory or intrathoracic, bone and soft-tissue tumors, lymphoma, and myeloma—point estimates remained directionally protective but statistically imprecise, reflecting limited precision after stratification rather than an absence of effect.

Characteristic	Bivalent non-recipients Events/Total (%)	Bivalent recipients Events/Total (%)	Pooled logistic HR (95% CI)	Effectiveness (%)
Primary Analysis				
All	5,391/64,353 (8.38)	1,068/20,395 (5.24)	0.778 (0.718-0.842)	22.2 (15.8-28.2)*
Subgroup Analysis				
(A) Age				
20-49	1,682/19,217 (8.75)	49/1,269 (3.86)	0.597 (0.437-0.816)	40.3 (18.4-56.3)*
50-64	1,805/25,143 (7.18)	209/4,885 (4.28)	0.773 (0.659-0.906)	22.7 (9.4-34.1)*
≥65	1,904/19,993 (9.52)	810/14,241 (5.69)	0.811 (0.736-0.895)	18.9 (10.5-26.4)*
(B) Previous vaccination state				
No	643/5,384 (11.94)			
Primary series only	1,009/13,202 (7.64)	52/968 (5.37)	0.856 (0.623-1.176)	14.4 (-17.6-37.7)
Booster one	2,731/34,441 (7.93)	346/7,201 (4.80)	0.748 (0.661-0.845)	25.2 (15.5-33.9)*
Booster two	1,008/11,326 (8.90)	670/12,226 (5.48)	0.789 (0.705-0.883)	21.1 (11.7-29.5)*
(C) Number of previous SARS-CoV2 infection				
1	5,294/62,226 (8.51)	1,027/19,769 (5.20)	0.761 (0.701-0.825)	23.9 (17.5-29.9)*
≥2	97/2,127 (4.56)	41/626 (6.55)	1.498 (0.973-2.306)	-49.8 (-130.6-2.7)
(D) Time since the most recent infection				
< 90	354/13,600 (2.60)	101/4,090 (2.47)	1.030 (0.790-1.342)	-3.0 (-34.2 - 21.0)
< 180	450/6,575 (6.84)	109/2,471 (4.41)	0.783 (0.615-0.997)	21.7 (0.3-38.5)*
< 1 year	4,252/41,865 (10.16)	802/13,112 (6.12)	0.732 (0.670-0.801)	26.8 (19.9-33.0)*
≥ 1 year	335/2,313 (14.48)	56/722 (7.76)	0.642 (0.458-0.899)	35.8 (10.1-54.2)*
(E) Cancer type				
Solid	5,139/61,630 (8.34)	1,036/19,713 (5.26)	0.781 (0.721-0.847)	21.9 (15.3-27.9)*
Hematologic	312/3,141 (9.93)	49/834 (5.88)	0.852 (0.597-1.217)	14.8 (-21.7-40.3)
(F) Cancer subtype				
Lip, oral cavity, pharynx	71/940 (7.55)	20/340 (5.88)	1.128 (0.618-2.059)	-12.8 (-105.9-38.2)
Colorectal	558/6,507 (8.58)	133/2,667 (4.99)	0.671 (0.534-0.843)	32.9 (15.7-46.6)*
Non-colorectal gastrointestinal	838/11,070 (7.57)	258/5,078 (5.08)	0.819 (0.692-0.970)	18.1 (3.0-30.8)*
Lung	334/3,420 (9.77)	97/1,711 (5.67)	0.643 (0.490-0.844)	35.7 (15.6-51.0)*
Respiratory / intrathoracic	56/713 (7.85)	12/264 (4.55)	0.583 (0.268-1.272)	41.7 (-27.2-73.2)
Bone/mesothelial/soft tissue	229/2,274 (10.07)	62/982 (6.31)	0.850 (0.616-1.197)	15.0 (-19.7-38.4)
Breast	929/11,442 (8.12)	104/2,202 (4.72)	0.715 (0.565-0.905)	28.5 (9.5-43.5)*
Gynecological	363/4,093 (8.87)	38/693 (5.48)	0.756 (0.515-1.110)	24.4 (-11.0-48.5)
Prostate	270/3,258 (8.29)	163/2,754 (5.92)	0.941 (0.744-1.189)	5.9 (-18.9-25.6)
Male urological	16/191 (8.38)	2/31 (6.45)	1.295 (0.305-5.494)	-29.5 (-449.4-69.5)
Urinary tract	273/3,347 (8.16)	103/1,614 (6.38)	0.979 (0.745-1.286)	2.1 (-28.6-25.5)
CNS	92/806 (11.41)	13/183 (7.10)	0.715 (0.357-1.433)	28.5 (-43.3-64.3)
Endocrine glands	1,606/18,831 (8.53)	169/3,318 (5.09)	0.760 (0.629-0.918)	24.0 (8.2-37.1)*
Lymphoma	149/1,442 (10.33)	31/418 (7.42)	1.339 (0.844-2.125)	-33.9 (-112.5-15.6)
Myeloma	113/923 (12.24)	20/262 (7.63)	0.780 (0.439-1.384)	22.0 (-38.4-56.1)
Leukemia	76/1,006 (7.55)	4/223 (1.79)	0.288 (0.094-0.883)	71.2 (11.7-90.6)*
Others	318/3,377 (9.42)	39/826 (4.72)	0.627 (0.423-0.930)	37.3 (7.0-57.7)*
(G) Chemotherapy				
No	5,138/61,043 (8.42)	1,016/19,481 (5.22)	0.773 (0.713-0.839)	23.9 (17.5-29.9)*
Yes	253/3,310 (7.64)	52/914 (5.69)	0.903 (0.623-1.307)	9.7 (-30.7-37.7)

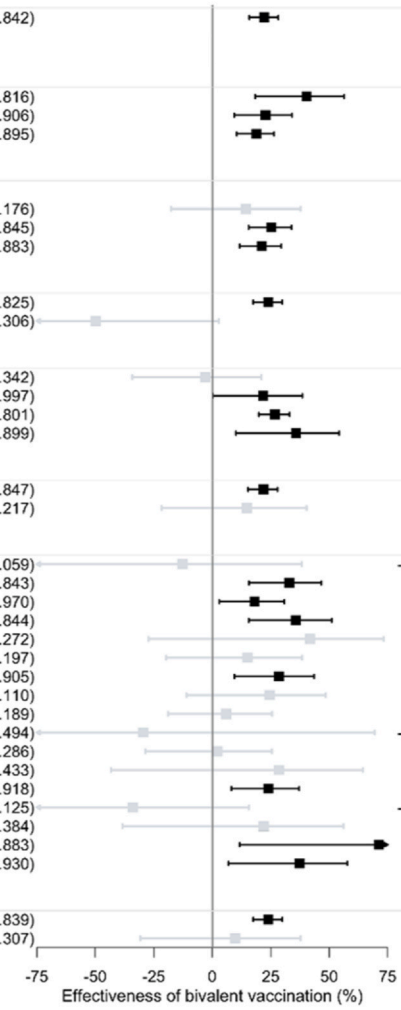

Fig. 3. Subgroup analysis of overall vaccine effectiveness. Subgroup analysis of the primary outcome is shown with the use of forest plot of the difference in percentage points and 95% confidence intervals, adjusted for predefined confounders. Time since the most recent infection and time since the most recent vaccination were calculated relative to the index date (October 4, 2022). Primary analysis adjusted for age, previous vaccination state, number of previous SARS-CoV-2 infections, time since the most recent infection, time since the most recent vaccination, cancer subtype, and chemotherapy; Adjustment sets differed by subgroup: (A) adjusted for previous vaccination state, number of previous infections, time since most recent infection, time since the most recent vaccination, cancer subtype, and chemotherapy; (B) adjusted for age, number of previous infections, time since most recent infection, time since the most recent vaccination, cancer subtype, and chemotherapy; (C) adjusted for age, previous vaccination state, time since most recent infection, time since the most recent vaccination, cancer subtype, and chemotherapy; (D) adjusted for age, previous vaccination state, time since the most recent vaccination number of previous infections, cancer subtype, and chemotherapy; (E) and (F) adjusted for age, previous vaccination state, number of previous SARS-CoV-2 infections, time since the most recent infection, time since the most recent vaccination and chemotherapy; (G) adjusted for age, previous vaccination state, number of previous infections, time since infection, and cancer subtype. Analyses in (E) were conducted after excluding "Others" (C76–C80, C97) from the cancer subtype classification, and stratified into solid and hematologic cancers. In contrast, (F) presents results for the full cancer subtype classification, including "Others," using the same adjustment set. Black markers and asterisks (*) indicate statistically significant vaccine protection; gray markers indicate non-significant estimates. Vaccine effectiveness (VE) = (1 − HR) × 100.

Discussion

In this nationwide TTE study of 84,748 adults with cancer and prior SARS-CoV-2 infection, bivalent COVID-19 vaccination was associated with a 22.2% reduction in SARS-CoV-2 reinfection risk. Reinfection rate was 5.2% in the bivalent vaccination group compared to 8.4% in the no-bivalent vaccination group. Protection, however, was heterogeneous across patient subgroups. Higher effectiveness was observed in younger adults, patients with solid tumors, and those with a more extensive prior vaccination history, including receipt of one or more monovalent boosters prior to bivalent vaccination. Furthermore, individuals with more distant prior infections also exhibited higher effectiveness. This finding likely reflects the waning of natural immunity over time, creating a larger immunologic margin for enhancement by the booster vaccination.

Our observed 22% reduction with bivalent vaccination reflects the incremental benefit of boosting in a population that had already acquired immunity through prior infection and vaccination, leaving a narrower margin for additional protection. This estimate is consistent with recent findings from Hong Kong, where an additional vaccine dose in the general population yielded a similarly modest reduction of 20%,[27] and contrasts with early studies reporting higher effectiveness of 45–60% against primary infection.[57,58] This pattern of attenuated but meaningful protection is biologically coherent and supports the continued value of bivalent vaccination in this population. Randomized trials, while establishing the foundational efficacy of COVID-19 vaccines, systematically excluded individuals with active malignancies.[59,60] Concurrently, immunogenicity studies have documented attenuated, albeit clinically meaningful immune responses, thereby supporting the biological plausibility of vaccine-

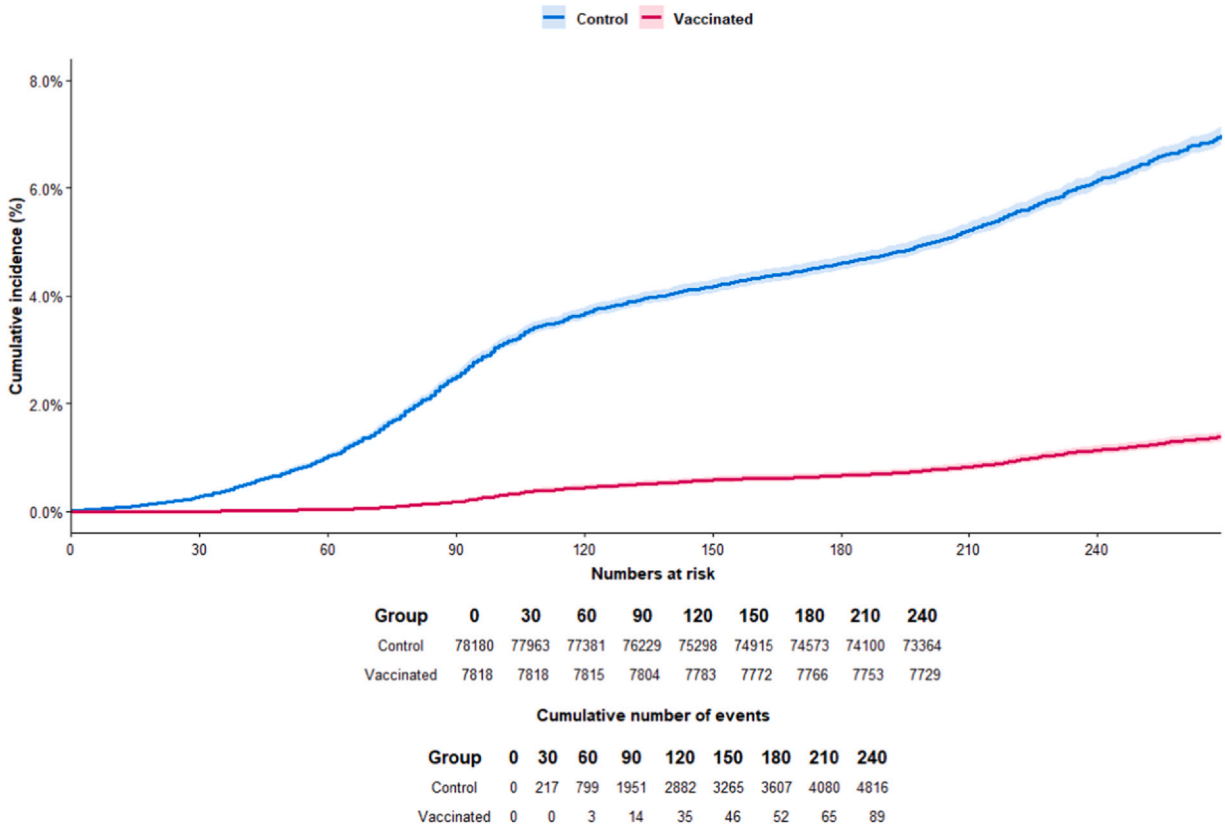

Fig. 4. Cumulative incidence of SARS-CoV-2 reinfection. Cumulative incidence of SARS-CoV-2 reinfection between vaccinated and control groups. The solid lines represent the estimated cumulative hazards, while the shaded areas indicate the 95% confidence intervals.

induced protection in this population.[61,62] Existing observational studies have further contributed valuable insights,[63–65] though estimates from designs that adequately account for time-varying confounding and adherence-related biases remain relatively limited.[65,66] Taken together, the present findings extend and strengthen the existing evidence base, offering population-level estimates that are consistent with and complementary to prior evidence.

Beyond this overall estimate, however, protection varied substantially according to host immune capacity. Age-related immunologic function and the timing of prior infections substantially shaped booster responsiveness. Younger adults and those with extensive vaccination histories experienced greater protection, consistent with established mechanisms of immunosenescence in which age-related declines in humoral and cellular immunity dampen vaccine responses.[67–69] The temporal dynamics of hybrid immunity similarly shaped effectiveness: individuals with recent infections derived minimal additional benefit, whereas those with more distant infection (> 1 year) showed substantially greater protection, consistent with the "ceiling effect" described in the general population.[70,71] Slightly reduced effectiveness among individuals with two or more prior boosters may reflect immune imprinting,[72] although alternative explanations—including residual confounding and imprecision—remain possible. Nevertheless, our findings align with those of previous studies identifying enhanced booster responsiveness among immunocompromised individuals.[23,73,74] In contrast, individuals with ≥2 prior infections had highly variable estimates, likely reflecting both immunologic vulnerability and statistical instability. Thus, host immune capacity—not merely cancer diagnosis—remains the key determinant of responsiveness to booster vaccination.

Tumor type further influenced vaccine effectiveness through heterogeneity in cancer-associated immune impairment. Attenuated protection in hematologic malignancies is consistent with prior studies showing that B-cell depleting therapies and profound lymphopenia diminish vaccine-induced antibody responses.[75–77] In contrast, the stronger protection observed in solid tumors—particularly colorectal, non-colorectal gastrointestinal, lung, and endocrine gland cancers—may reflect relatively preserved immune function in these populations, including the capacity for effective memory B-cell mediated recall responses,[78] although the exact mechanisms cannot be fully delineated from our data. These findings collectively highlight the diverse immune landscapes across malignancies and suggest that patients with severe B-cell impairment may require enhanced or alternative preventive strategies.

Bivalent vaccination has important clinical implications for optimizing cancer care during the endemic phase. The imperative for vaccination in cancer patients extends beyond preventing temporary illness to maintaining the continuity of cancer care. Unlike the general population, SARS-CoV-2 infections in cancer patients can disrupt oncology treatment schedules, often necessitating temporary delays or modifications that may compromise the dose intensity for time-sensitive regimens. A reinfection reduction of approximately 22% overall—and point estimates approaching 35–40% in immune-competent subgroups—suggests that bivalent vaccination can meaningfully reduce oncologic disruption during the endemic phase.

As national immunization programs transition toward routine seasonal vaccination, our findings provide empirical evidence to inform whether cancer patients should continue to receive COVID-19 vaccination. The reduction in reinfection risk, combined with consistent protection across key clinical subgroups, supports the continued prioritization of COVID-19 vaccination for patients with cancer. However, the substantial heterogeneity in vaccine effectiveness across clinical subgroups highlights the need to shift from a uniform policy to a more tailored approach. For patients with relatively preserved immune function, such as those with solid tumors,

routine seasonal boosting remains a highly effective strategy. Conversely, for highly vulnerable subgroups exhibiting attenuated protection, such as those with hematologic malignancies, routine vaccination alone may be insufficient, warranting individualized vaccine timing in relation to anticancer therapy or exploring modified booster schedules.[79] For these highest-risk patients, vaccination must be complemented by broader preventive measures. This includes ensuring prompt access to early antiviral therapies upon infection in high-risk immunocompromised patients,[80,81] alongside sustained adherence to targeted infection control practices. Currently, empirical evidence defining the optimal combination of these alternative approaches remains sparse, and establishing comprehensive, evidence-based preventive protocols for patients with severely impaired vaccine responses represents an important priority for future research.

Strengths and limitations

This study provides solid methodological and clinical advancements in evaluating vaccine effectiveness for vulnerable populations. By leveraging TTE principles using the CCW approach and integrating nationwide clinical and vaccination data, we explicitly address critical methodological challenges, including immortal time bias, adherence-related censoring, and time-varying confounding, thereby strengthening the basis for causal inference relative to conventional observational approaches under stated assumptions. Clinically, this integration of the nationwide K-CoV-N cohort enables highly generalizable, population-based estimates for cancer patients, who are systematically excluded from randomized trials. Furthermore, to our knowledge, this is one of the first nationwide studies to have applied a TTE framework to evaluate bivalent booster effectiveness against reinfection in cancer patients, by comprehensively characterizing the heterogeneity of protection across diverse tumor subtypes, distinct immune histories, and hybrid immunity profiles. By focusing specifically on reinfection during the endemic phase, our findings provide timely and granular evidence crucial for mitigating risk in the high-risk population. While TTE does not eliminate the need for untestable assumptions, it contributes a rigorously designed observational estimate that, alongside immunogenicity studies and prior effectiveness analyses, advances the triangulation of causal understanding in this population.

As with any observational study, however, these findings should be interpreted in light of several limitations. First, the lack of information on some unmeasured confounders, such as health-seeking behavior, cancer severity, treatment cycle, and treatment intensity, could lead to residual confounding. Although we adjusted for multiple proxies (chemotherapy, CCI, and cancer subtype), residual confounding by indication may persist. Second, per-protocol censoring could introduce bias if adherence was related to unmeasured factors. We addressed this using IPCW, although some residual bias is unavoidable. Third, as with all TTE analyses, the validity of our estimates relies on assumptions that cannot be fully verified empirically, including consistency of the bivalent vaccine as a well-defined intervention and structural positivity in subgroups with near-universal or near-zero uptake. Fourth, reliance on clinical testing may have resulted in underascertainment due to increased at-home testing. However, given the similarly high healthcare contact in both groups, any misclassification was likely nondifferential, biasing estimates toward the null. Finally, follow-up ended in June 2023, limiting the evaluation of long-term waning and effectiveness against later Omicron sublineages.

Conclusions

In this nationwide target trial emulation of 84,748 cancer patients, bivalent COVID-19 vaccination reduced SARS-CoV-2 reinfection risk by 22%, albeit with substantial heterogeneity. Protection was greater among individuals with relatively preserved immune function, including younger adults and those with a more extensive prior vaccination history such as receipt of one or more monovalent boosters, and those with waning natural immunity. These findings highlight the continued clinical value of seasonal boosting in the oncology population, particularly to restore protection as natural immunity declines, while underscoring that a uniform vaccination policy is insufficient. As the pandemic transitions to the endemic phase, optimizing protection for cancer patients will require tailored strategies that account for tumor type, treatment context, and immune history, integrating vaccination as a cornerstone, not a ceiling, of prevention in this vulnerable population. In an era when SARS-CoV-2 exposure is inevitable but reinfection need not be, these findings offer a framework for developing more precise and individualized preventive strategies for cancer populations, one that builds on an established evidence base, contributes population-level effectiveness estimates in a historically underrepresented population, and positions seasonal vaccination not as a policy uncertainty to be resolved, but as a foundation to be optimized.

CRediT authorship contribution statement

Concept and design: H.K.Kim, S.Jo, Acquisition, analysis, or interpretation of data: All authors, Drafting of the manuscript: H.K.Kim, Critical revision of the manuscript for important intellectual content: All authors, Statistical analysis: H.K.Kim, Obtained funding: S.I.Cho, Administrative, technical, or material support: H.K.Kim, S.Jo, Supervision: S.Jo, S.I.Cho.

Ethical approval

This study was approved from review by the Institutional Review Board of Seoul National University (IRB No. E2503/003-003).

Funding

This research was supported by the Bio & Medical Technology Development Program of the National Research Foundation, funded by the Korean Government (No. 2021M3E5E3081366), and by the National Research Foundation of Korea through the BK21 Center for Integrative Response to Health Diasters, Graduate School of Public Health, Seoul National University (No.4199990514025). The funding sources were not involved in the study design or writing and editing of this report.

Data availability

The data used in this study are owned by the Korea Disease Control and Prevention Agency and the National Health Insurance Service and cannot be shared publicly due to legal and privacy restrictions. Access to the data requires approval from the respective data custodians. Statistical code used for the analyses is available from the corresponding author upon reasonable request

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used an artificial intelligence–based language model (ChatGPT, GPT-5.1; OpenAI) to

assist with language editing and text refinement. After using this tool, the authors reviewed and edited all content and take full responsibility for the integrity and accuracy of the manuscript.

Acknowledgements

This study used the database of the KDCA (Korea Disease Control and Prevention Agency, Republic of Korea) and the NHIS (National Health Insurance Service, Republic of Korea) for policy and academic research. The research numbers of this study are KDCA-NHIS-2024-11-070.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jinf.2026.106753.

References

1. Centers for Disease Control and Prevention. COVID-19 vaccination guidance forpeople who are immunocompromised. Covid; 2025. ⟨⟨https://www.cdc.gov/covid/hcp/vaccine-considerations/immunocompromised.html⟩⟩ [Accessed November 12,2025].
2. Korea Disease Control and Prevention Agency. COVID-19 national immunizationprogram (temporary implementation). ⟨⟨https://nip.kdca.go.kr/irhp/covd/goCov19Vcnt.do?menuLv=4&menuCd=47⟩⟩ [Accessed November 12, 2025].
3. Giesen N, Sprute R, Rüthrich M, Khodamoradi Y, Mellinghoff SC, Beutel G, et al. *2021 update of the AGIHO guideline onevidence-based management of COVID-19 in patients with cancer regardingdiagnostics, viral shedding, vaccination and therapy. Eur J Cancer* 2021;**147**:154–60. https://doi.org/10.1016/j.ejca.2021.01.033
4. Fendler A, de Vries EGE, GeurtsvanKessel CH, Haanen JB, Wörmann B, Turajlic S, et al. *COVID-19 vaccines inpatients with cancer: immunogenicity, efficacy and safety. Nat Rev Clin Oncol* 2022;**19**(6):385–401. https://doi.org/10.1038/s41571-022-00610-8
5. Salunke AA, Nandy K, Pathak SK, Shah J, Kamani M, Kottakota V, et al. *Impact of COVID-19 in cancer patientson severity of disease and fatal outcomes. a systematic review and meta-analysis,Diabetes Metab Syndr Clin Res Rev* 2020;**14**(5):1431–7. https://doi.org/10.1016/j.dsx.2020.07.037
6. Chavez-MacGregor M, Lei X, Zhao H, Scheet P, Giordano SH. *Evaluation ofCOVID-19 mortality and adverse outcomes in US patients with or without cancer. JAMA Oncol* 2022;**8**(1):69–78. https://doi.org/10.1001/jamaoncol.2021.5148
7. Aurer I, Moss P, Goldman M, Tuthill M, Einsele H, Casañas C, Comabella i, et al. *COVID-19 burden of illness in people whoare immunocompromised due to cancer: an expert opinion review.* oyaf074 *Oncologist* 2025;**30**(6). https://doi.org/10.1093/oncolo/oyaf074
8. National Cancer Institute. COVID-19 vaccines and people with cancer - NCI; 2021. ⟨⟨https://www.cancer.gov/about-cancer/coronavirus/covid-19-vaccines-people-with-cancer⟩⟩ [Accessed November 12, 2025].
9. Jing X, Wang M, Lu S, Niu J, Chen F, Yin H, et al. *Evaluating the impact of COVID-19 on cancer care:a comprehensive analysis of treatment modifications, risk factors, and patientoutcomes. BMC Infect Dis* 2025;**25**(1):779. https://doi.org/10.1186/s12879-025-11172-2
10. Shah R, Hanna NM, Loo CE, David M, Mafra A, Fink H, et al. *The global impact of the COVID-19pandemic on delays and disruptions in cancer care services: a systematic review andmeta-analysis. Nat Cancer* 2025;**6**(1):194–204. https://doi.org/10.1038/s43018-024-00880-4
11. Ehmsen S, Asmussen A, Jeppesen SS, Nilsson AC, Kragh A, Frederiksen H, et al. *Increased antibody titers andreduced seronegativity following fourth mRNA COVID-19 vaccination in patientswith cancer. Cancer Cell* 2022;**40**(8):800–1. https://doi.org/10.1016/j.ccell.2022.06.012
12. Breathnach AS, Duncan CJA, Bouzidi KE, Hanrath AT, Payne BAI, Randell PA, et al. *Prior COVID-19 protectsagainst reinfection, even in the absence of detectable antibodies. J Infect* 2021;**83**(2):237–79. https://doi.org/10.1016/j.jinf.2021.05.024
13. Slezak J, Bruxvoort K, Fischer H, Broder B, Ackerson B, Tartof S. *Rate andseverity of suspected SARS-Cov-2 reinfection in a cohort of PCR-positive COVID-19patients.* 1860.e7-1860.e71860.e10 *Clin Microbiol Infect* 2021;**27**(12). https://doi.org/10.1016/j.cmi.2021.07.030
14. Kim JY, Chang E, Jang HM, Cha JH, Son JY, Jang CY, et al. *Reinfection of SARS-CoV-2 variants inimmunocompromised patients with prolonged or relapsed viral shedding. InfectChemother* 2025;**57**(1):81–92. https://doi.org/10.3947/ic.2024.0098
15. Rosenblum HG, Wallace M, Godfrey M, Roper LE, Hall E, Fleming-Dutra KE, et al. *Interim recommendations fromthe advisory committee on immunization practices for the use of bivalent boosterdoses of COVID-19 vaccines - United States, October 2022. MMWR Morb Mortal WklyRep* 2022;**71**(45):1436–41. https://doi.org/10.15585/mmwr.mm7145a2
16. World Health Organization. Interim statement on decision-making considerationsfor the use of variant updated COVID-19 vaccines. ⟨⟨https://www.who.int/news/item/17–06-2022-interim-statement-on-decision-making-considerations-for-the-use-of-variant-updated-covid-19-vaccines⟩⟩ [Accessed April 2, 2026].
17. Wang Q, Bowen A, Valdez R, Gherasim C, Gordon A, Liu L, et al. *Antibody response to Omicron BA.4–BA.5bivalent booster. N Engl J Med* 2023;**388**(6):567–9. https://doi.org/10.1056/NEJMc2213907
18. Chalkias S, Whatley JL, Eder F, Essink B, Khetan S, Bradley P, et al. *Original SARS-CoV-2 monovalent andOmicron BA.4/BA.5 bivalent COVID-19 mRNA vaccines: phase 2/3 trial interimresults. Nat Med* 2023;**29**(9):2325–33. https://doi.org/10.1038/s41591-023-02517-y
19. Arbel R, Peretz A, Sergienko R, Friger M, Beckenstein T, Duskin-Bitan H, et al. *Effectiveness of a bivalent mRNA vaccinebooster dose to prevent severe COVID-19 outcomes: a retrospective cohort study. Lancet Infect Dis* 2023;**23**(8):914–21. https://doi.org/10.1016/S1473-3099(23)00122-6
20. Skarbinski J, Elkin EP, Ziemba YC, Kazemian E, Wilson BM, Siddiqui H, et al. *COVID-19 vaccine booster uptake andeffectiveness among US adults with cancer. JAMA Oncol* 2025;**11**(9):999–1010. https://doi.org/10.1001/jamaoncol.2025.2020
21. Ogata T, Hattori S, Kawachi K, Ishikane M, Tanaka H. *Durability ofeffectiveness of a booster dose of COVID-19 bivalent (ancestral/BA.4-5) vaccineagainst all-cause mortality in older adults in Japan, October 2022 to September2023. Vaccine* 2025;**65**:127796. https://doi.org/10.1016/j.vaccine.2025.127796
22. Lazar Neto F, Mercadé-Besora N, Raventós B, Pérez-Crespo L, Castro Junior G, Ranzani OT, et al. *Effectiveness of COVID-19vaccines against severe COVID-19 among patients with cancer in Catalonia, Spain. Nat Commun* 2024;**15**(1):5088. https://doi.org/10.1038/s41467-024-49285-y
23. Hall V, Foulkes S, Insalata F, Kirwan P, Saei A, Atti A, et al. *Protection against SARS-CoV-2 after Covid-19 vaccination and previous infection. N Engl J Med* 2022;**386**(13):1207–20. https://doi.org/10.1056/NEJMoa2118691
24. Hammerman A, Sergienko R, Friger M, Beckenstein T, Peretz A, Netzer D, et al. *Effectiveness of the BNT162b2vaccine after recovery from Covid-19. N Engl J Med* 2022;**386**(13):1221–9. https://doi.org/10.1056/NEJMoa2119497
25. Nielsen KF, Moustsen-Helms IR, Schelde AB, Gram MA, Emborg HD, Nielsen J, et al. *Vaccine effectivenessagainst SARS-CoV-2 reinfection during periods of Alpha, Delta, or Omicrondominance: a Danish nationwide study. PLOS Med* 2022;**19**(11):e1004037. https://doi.org/10.1371/journal.pmed.1004037
26. Lewis N, Chambers LC, Chu HT, Fortnam T, De Vito R, Gargano LM, et al. *Effectiveness associated withvaccination after COVID-19 recovery in preventing reinfection. JAMA Netw Open* 2022;**5**(7):e2223917. https://doi.org/10.1001/jamanetworkopen.2022.23917
27. Lin W, Kung KH, Chan CL, Chuang SK, Au KW. *Characteristics and riskfactors associated with COVID-19 reinfection in Hong Kong: a retrospective cohortstudy. Epidemiol Infect* 2025;**153**:e30. https://doi.org/10.1017/S0950268825000172
28. Tan WC, Tan JYJ, Lim JSJ, Tan RYC, Lee ARYB, Leong FL, et al. *COVID-19 severity and waning immunityafter up to 4 mRNA vaccine doses in 73 608 patients with cancer and 621 475matched controls in Singapore: a nationwide cohort study. JAMA Oncol* 2023;**9**(9):1221–9. https://doi.org/10.1001/jamaoncol.2023.2271
29. Wu JTY, La J, Branch-Elliman W, Huhmann LB, Han SS, Parmigiani G, et al. *Association of COVID-19 vaccinationwith SARS-CoV-2 infection in patients with cancer. JAMA Oncol* 2022;**8**(2):1–6. https://doi.org/10.1001/jamaoncol.2021.5771
30. Lee LYW, Ionescu MC, Starkey T, Little M, Tilby M, Tripathy AR, et al. *COVID-19: third dose booster vaccineeffectiveness against breakthrough coronavirus infection, hospitalisations and deathin patients with cancer: a population-based study. Eur J Cancer* 2022;**175**:1–10. https://doi.org/10.1016/j.ejca.2022.06.038
31. Hernán MA, Sauer BC, Hernández-Díaz S, Platt R, Shrier I. *Specifying atarget trial prevents immortal time bias and other self-inflicted injuries inobservational analyses. J Clin Epidemiol* 2016;**79**:70–5. https://doi.org/10.1016/j.jclinepi.2016.04.014
32. Hernán MA, Wang W, Leaf DE. *Target trial emulation: a framework forcausal inference from observational data. JAMA* 2022;**328**(24):2446–7. https://doi.org/10.1001/jama.2022.21383
33. Hwang H, Lee W, Ahn S, Choi YS, Kwon SL, Lee D, et al. *Implementation plan for the coronavirus disease2019 vaccination for the 2024–2025 season: recommendations of the 6th expertcommittee on immunization practices. Public Health Wkly Rep* 2025;**18**(2):90–102. https://doi.org/10.56786/PHWR.2025.18.2.3
34. Centers for Disease Control and Prevention. 2025–2026 COVID-19 vaccinationguidance. ⟨⟨https://www.cdc.gov/covid/hcp/vaccine-considerations/routine-guidance.html⟩⟩ [Accessed November 12, 2025].
35. Ministry of Health and Welfare. National Health Insurance: population coverage, contribution rate & benefit payment < health statistics. ⟨⟨https://www.mohw.go.kr/menu.es?mid=a20310000000⟩⟩ [Accessed November 12, 2025].
36. Song SO, Jung CH, Song YD, Park CY, Kwon HS, Cha BS, et al. *Background and data configuration processof a nationwide population-based study using the Korean national health insurancesystem. Diabetes Metab J* 2014;**38**(5):395–403. https://doi.org/10.4093/dmj.2014.38.5.395
37. Lee H, Choi S, Eom E, Song GH, Kim SS. *Introduction to the researchresults of coronavirus disease 2019 big data (K-COV-N) and the correspondingutilization plan. Public Health Wkly Rep* 2024;**17**(48):2147–59. https://doi.org/10.56786/PHWR.2024.17.48.4
38. Lee HE, Jeong NY, Park M, Lim E, Kim JA, Won H, et al. *Effectiveness of COVID-19 vaccines againstsevere outcomes in cancer patients: real-world evidence from self-controlled riskinterval and retrospective cohort studies. J Infect Public Health* 2024;**17**(5):854–61. https://doi.org/10.1016/j.jiph.2024.03.015
39. Jang Y, Kim T, Kim BHS, Kim JH, Seong H, Kim YJ, et al. *Economic burden of cancer for the first fiveyears after cancer diagnosis in patients with human immunodeficiency virus inKorea. J Cancer Prev* 2023;**28**(2):53–63. https://doi.org/10.15430/JCP.2023.28.2.53
40. Kim RK, Choe YJ, Jang EJ, Hwang JH, Lee KH, Shim JA, et al. *Comparative effectiveness of covid-19bivalent versus monovalent mRNA vaccines in the early stage of bivalent vaccinationin Korea: October 2022 to January 2023. J Korean Med Sci* 2023;**38**(46). https://doi.org/10.3346/jkms.2023.38.e396

41. Hernán MA, Robins JM. *Using big data to emulate a target trial when arandomized trial is not available. Am J Epidemiol* 2016;**183**(8):758–64. https://doi.org/10.1093/aje/kwv254
42. Reep CAT, Wils EJ, Heunks L. *Opportunities, challenges and futureperspectives for target trial emulation in critical care clinical research. Crit Care* 2025;**29**(1):484. https://doi.org/10.1186/s13054-025-05723-x
43. Riggi E, Segelov E, Di Tanna GL. *Target trial emulation: bridgingobservational studies and randomized trials for health decision-making. J Comp EffRes* 2026;**15**(4):e250180. https://doi.org/10.57264/cer-2025-0180
44. Cashin AG, Hansford HJ, Hernán MA, Swanson SA, Lee H, Jones MD, et al. *Transparent reporting ofobservational studies emulating a target trial-the TARGET statement. JAMA* 2025;**334**(12):1084–93. https://doi.org/10.1001/jama.2025.13350
45. Pfizer-BioNTech, Pfizer-BioNTech COVID-19 vaccine (BNT162, PF-07302048)vaccines and related biological products advisory committee briefing document, 2020, U.S. Food and Drug Administration; Silver Spring (MD).
46. Yahav D, Yelin D, Eckerle I, Eberhardt CS, Wang J, Cao B, et al. *Definitions for coronavirus disease 2019reinfection, relapse and PCR re-positivity. Clin Microbiol Infect* 2021;**27**(3):315–8. https://doi.org/10.1016/j.cmi.2020.11.028
47. Centers for Disease Control and Prevention (CDC). About reinfection. COVID-19;2025. 〈〈https://www.cdc.gov/covid/about/reinfection.html〉〉 [Accessed November12, 2025].
48. Levin EG, Lustig Y, Cohen C, Fluss R, Indenbaum V, Amit S, et al. *Waning immune humoral response toBNT162b2 Covid-19 vaccine over 6 months. N Engl J Med* 2021;**385**(24):e84. https://doi.org/10.1056/NEJMoa2114583
49. Ruff J, Zhang Y, Kappel M, Rathi S, Watkins K, Zhang L, et al. *Rapid increase in suspected SARS-CoV-2reinfections, Clark County, Nevada, USA, december 2021. Emerg Infect Dis* 2022;**28**(10):1977–81. https://doi.org/10.3201/eid2810.221045
50. Muehleman D, Gruner B, Hogan V, Fanning P, Garrett C, Meyer J, et al. *Using SARS-CoV-2 sequencing data toidentify reinfection cases in the global emerging infections surveillance program,United States. Emerg Infect Dis* 2024;**30**(2):S53–61. https://doi.org/10.3201/eid3014.240231
51. Hernán MA, Dahabreh IJ, Dickerman BA, Swanson SA. *The target trialframework for causal inference from observational data: why and when is it helpful? Ann Intern Med* 2025;**178**(3):402–7. https://doi.org/10.7326/ANNALS-24-01871
52. Hernán MA, Sterne JAC, Higgins JPT, Shrier I, Hernández-Díaz S. *Astructural description of biases that generate immortal time. Epidemiology* 2025;**36**(1):107–14. https://doi.org/10.1097/EDE.0000000000001808
53. Gaber CE, Hanson KA, Kim S, Lund JL, Lee TA, Murray EJ. *The clone-censor-weight method in pharmacoepidemiologic research: foundations andmethodological implementation. Curr Epidemiol Rep* 2024;**11**(3):164–74. https://doi.org/10.1007/s40471-024-00346-2
54. Cole SR, Hernán MA. *Adjusted survival curves with inverse probabilityweights. Comput Methods Prog Biomed* 2004;**75**(1):45–9. https://doi.org/10.1016/j.cmpb.2003.10.004
55. Hernán MA, Brumback B, Robins JM. *Marginal structural models toestimate the joint causal effect of nonrandomized treatments. J Am Stat Assoc* 2001;**96**(454):440–8. https://doi.org/10.1198/016214501753168154
56. Hernán MÁ, Brumback B, Robins JM. *Marginal structural models toestimate the causal effect of zidovudine on the survival of HIV-positive men. Epidemiology* 2000;**11**(5):561–70. https://doi.org/10.1097/00001648-200009000-00012
57. Lee LYW, Starkey T, Ionescu MC, Little M, Tilby M, Tripathy AR, et al. *Vaccine effectiveness against COVID-19 breakthrough infections in patients with cancer (UKCCEP): a population-basedtest-negative case-control study. Lancet Oncol* 2022;**23**(6):748–57. https://doi.org/10.1016/S1470-2045(22)00202-9
58. Lee LYW, Ionescu MC, Starkey T, Little M, Tilby M, Tripathy AR, et al. *COVID-19: third dose booster vaccineeffectiveness against breakthrough coronavirus infection, hospitalisations and deathin patients with cancer: a population-based study. Eur J Cancer* 2022;**175**:1–10. https://doi.org/10.1016/j.ejca.2022.06.038
59. Hyer RN. *Insights for oncology trials garnered from the rapid development of anmRNA COVID-19 vaccine. Cancer J* 2022;**28**(2):146–50. https://doi.org/10.1097/PPO.0000000000000587
60. Thomas SJ, Perez JL, Lockhart SP, Hariharan S, Kitchin N, Bailey R, et al. *Efficacy and safety of the BNT162b2mRNA COVID-19 vaccine in participants with a history of cancer: subgroup analysisof a global phase 3 randomized clinical trial. Vaccine* 2022;**40**(10):1483–92. https://doi.org/10.1016/j.vaccine.2021.12.046
61. Cavanna L, Citterio C, Toscani I. *COVID-19 vaccines in cancer patients.Seropositivity and safety. Systematic review and meta-analysis. Vaccines* 2021;**9**(9):1048. https://doi.org/10.3390/vaccines9091048
62. Becerril-Gaitan A, Vaca-Cartagena BF, Ferrigno AS, Mesa-Chavez F, Barrientos-Gutiérrez T, Tagliamento M, et al. *Immunogenicity andrisk of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) infectionafter coronavirus disease 2019 (COVID-19) vaccination in patients with cancer: asystematic review and meta-analysis. Eur J Cancer* 2022;**160**:243–60. https://doi.org/10.1016/j.ejca.2021.10.014
63. Ruiz JI, Lopez-Olivo MA, Geng Y, Suarez-Almazor ME. *COVID-19vaccination in patients with cancer receiving immune checkpoint inhibitors: asystematic review and meta-analysis. J Immunother Cancer* 2023;**11**(2):e006246. https://doi.org/10.1136/jitc-2022-006246
64. Mekkawi R, Elkattan BA, Shablak A, Bakr M, Yassin MA, Omar NE. *COVID-19 vaccination in cancer patients: a review article.* 10732748221106266 *Cancer Control* 2022;**29**. https://doi.org/10.1177/10732748221106266
65. Galmiche S, Luong Nguyen LB, Tartour E, de Lamballerie X, Wittkop L, Loubet P, et al. *Immunological and clinicalefficacy of COVID-19 vaccines in immunocompromised populations: a systematic review. Clin Microbiol Infect* 2022;**28**(2):163–77. https://doi.org/10.1016/j.cmi.2021.09.036
66. Kuderer NM, Lyman GH. *COVID-19 vaccine effectiveness in patients withcancer: remaining vulnerabilities and uncertainties. Lancet Oncol* 2022;**23**(6):693–5. https://doi.org/10.1016/S1470-2045(22)00252-2
67. Han S, Georgiev P, Ringel AE, Sharpe AH, Haigis MC. *Age-associatedremodeling of T cell immunity and metabolism. Cell Metab* 2023;**35**(1):36–55. https://doi.org/10.1016/j.cmet.2022.11.005
68. Lazuardi L, Jenewein B, Wolf AM, Pfister G, Tzankov A, Grubeck-Loebenstein B. *Age-related loss of naïve T cells and dysregulation of T-cell/B-cellinteractions in human lymph nodes. Immunology* 2005;**114**(1):37–43. https://doi.org/10.1111/j.1365-2567.2004.02006.x
69. Foster WS, Marcial-Juárez E, Linterman MA. *The cellular factors thatimpair the germinal center in advanced age. J Immunol* 2025;**214**(5):862–71. https://doi.org/10.1093/jimmun/vkae039
70. Nilles EJ, Roberts K. *M. d e S. Aubin, H. Mayfield, A.C. Restrepo, S. Garnier, et al., Convergence of SARS-CoV-2 spikeantibody levels to a population immune setpoint. eBioMedicine* 2024;**108**. https://doi.org/10.1016/j.ebiom.2024.105319
71. Srivastava K, Carreño JM, Gleason C, Monahan B, Singh G, Abbad A, et al. *SARS-CoV-2-infection- and vaccine-induced antibody responses are long lasting with an initial waning phase followedby a stabilization phase.* e4 *Immunity* 2024;**57**(3):587–99. https://doi.org/10.1016/j.immuni.2024.01.017
72. Wheatley AK, Fox A, Tan HX, Juno JA, Davenport MP, Subbarao K, et al. *Immune imprinting and SARS-CoV-2vaccine design. Trends Immunol* 2021;**42**(11):956–9. https://doi.org/10.1016/j.it.2021.09.001
73. Wang J, Tong Y, Li D, Li J, Li Y. *The impact of age difference on the efficacyand safety of COVID-19 vaccines: a systematic review and meta-analysis. FrontImmunol* 2021;**12**:758294. https://doi.org/10.3389/fimmu.2021.758294
74. Lin DY, Gu Y, Xu Y, Wheeler B, Young H, Sunny SK, et al. *Association of primary and booster vaccination andprior infection with SARS-CoV-2 infection and severe COVID-19 outcomes. JAMA* 2022;**328**(14):1415–26. https://doi.org/10.1001/jama.2022.17876
75. Hofsink Q, Lissenberg-Witte BI, Haggenburg S, Goorhuis A, Hazenberg MD, Rutten CE, et al. *COVID-19 vaccine effectiveness in patients with hematologic malignancies: a nationwide cohort study. J Infect Dis* 2025;**233**(2):e363–72. https://doi.org/10.1093/infdis/jiaf523
76. Jullien M, Le Bourgeois A, Coste-Burel M, Peterlin P, Garnier A, Audrain M, et al. *Profound B-Cell lymphopenia isa major factor predicting poor humoral response after BNT162b2 mRNA Sars-Cov-2vaccines in ransplantation. Blood* 2021;**138**:3911. https://doi.org/10.1182/blood-2021-150439
77. Piechotta V, Mellinghoff SC, Hirsch C, Brinkmann A, Iannizzi C, Kreuzberger N, et al. *Effectiveness, immunogenicity,and safety of COVID-19 vaccines for individuals with hematological malignancies: asystematic review. Blood Cancer J* 2022;**12**(5):86. https://doi.org/10.1038/s41408-022-00684-8
78. Terreri S, Piano Mortari E, Vinci MR, Russo C, Alteri C, Albano C, et al. *Persistent B cell memory after SARS-CoV-2 vaccination is functional during breakthrough infections.* e4 *Cell Host Microbe* 2022;**30**(3):400–8. https://doi.org/10.1016/j.chom.2022.01.003
79. Kamboj M, Bohlke K, Baptiste DM, Dunleavy K, Fueger A, Jones L, et al. *Vaccination of adults with cancer:aSCO guideline. J Clin Oncol* 2024;**42**(14):1699–721. https://doi.org/10.1200/JCO.24.00032
80. Cesaro S, Ljungman P, Mikulska M, Hirsch HH, Navarro D, Cordonnier C, et al. *Post-pandemic recommendations forthe management of COVID-19 in patients with haematological malignancies orundergoing cellular therapy, from the European Conference on Infections inLeukaemia (ECIL-10). Leukemia* 2025;**39**(9):2061–71. https://doi.org/10.1038/s41375-025-02649-9
81. Piñana JL, Heras I, Aiello TF, García-Cadenas I, Vazquez L, Lopez-Jimenez J, et al. *Remdesivir or Nirmatrelvir/Ritonavirtherapy for Omicron SARS-CoV-2 infection in hematological patients and celltherapy recipients. Viruses* 2023;**15**(10):2066. https://doi.org/10.3390/v15102066